CS-471: Machine Learning

Week 7

Instructor: Dr. Hashir Kiani

Classroom Etiquette and Participation Guidelines

- Treat everyone with respect and courtesy at all times
- Speak only when it is your turn and raise your hand before contributing to the discussion
- Do not interrupt others while they are speaking
- Join the class on time
- Keep mobile phones on silent during the lecture

Lecture Outline

- Recap of Previous Lecture
- CART Example
- Decision Tree Pruning
- Ensemble Methods

Recap of Previous Lecture

- How does C4.5 handle missing values?

- How is regression done using CART Algorithm?

CART for Regression Example

Student	Study Hours	Attendance (%)	Marks
A	1	50	35
B	5	80	75
C	6	85	82
D	2	40	40
E	3	60	55
F	8	90	92

Introduction to Pruning in C4.5

- What is pruning?
  - A technique to reduce overfitting by removing unnecessary branches in the decision tree.
- Helps improve generalization to unseen data.
- Why prune?
  - Prevents overfitting to noise in training data.
  - Reduces tree complexity.
  - Improves model interpretability and efficiency.

Types of Pruning in Decision Trees

- Pre-Pruning (Early Stopping)
  - Stops tree growth before it becomes too complex.
  - Uses thresholds like max depth or min samples per node.

- Post-Pruning (Error-Based Pruning)
  - Grows full tree first, then removes unnecessary branches.
  - C4.5 uses error-based pruning to estimate classification error and remove ineffective branches.

How to detect irrelevant node

- A test node with high disorder might be testing an irrelevant attribute
- How high a disorder means irrelevant attribute?
- Can use a statistical significance test (Chi Squared Test)
- Null hypothesis is that the attribute is irrelevant
- Find the χ^2 value

χ^2 Pruning

- Suppose for a particular subset k, the actual number of positive examples is p_k and the actual number of negative examples is n_k.
- The expected number of positive and negative examples assuming the attribute is irrelevant are:

$$\hat{p}_k = p \times \frac{p_k + n_k}{p+n} \qquad \hat{n}_k = n \times \frac{p_k + n_k}{p+n}$$

Where p and n are the total number of positive and negative examples in that test

- The χ^2 value can then be found by:

$$\chi^2 = \sum_{k=1}^{d} \frac{(p_k - \hat{p}_k)^2}{\hat{p}_k} + \frac{(n_k - \hat{n}_k)^2}{\hat{n}_k}$$

Where d is the total number of subsets for that test

χ^2 Pruning

- Degrees of freedom is $d - 1$ where d is the total number of subsets

	Significance Level				
DF	0.1	0.05	0.02	0.01	0.001
1	2.706	3.841	5.412	6.635	10.827
2	4.605	5.991	7.824	9.210	13.815
3	6.251	7.815	9.837	11.345	16.268
4	7.779	9.488	11.668	13.277	18.465
5	9.236	11.070	13.388	15.086	20.517

- Find the critical value for relevant significance level and degrees of freedom from table
- If own value is below critical value, then attribute is irrelevant, and tree is pruned

How Post-Pruning Works in C4.5

- Step 1: Grow the Full Decision Tree
  - The tree is expanded to full depth based on Gain Ratio.

- Step 2: Compute Error for Each Node

$$Error(Node) = \frac{E}{N}$$

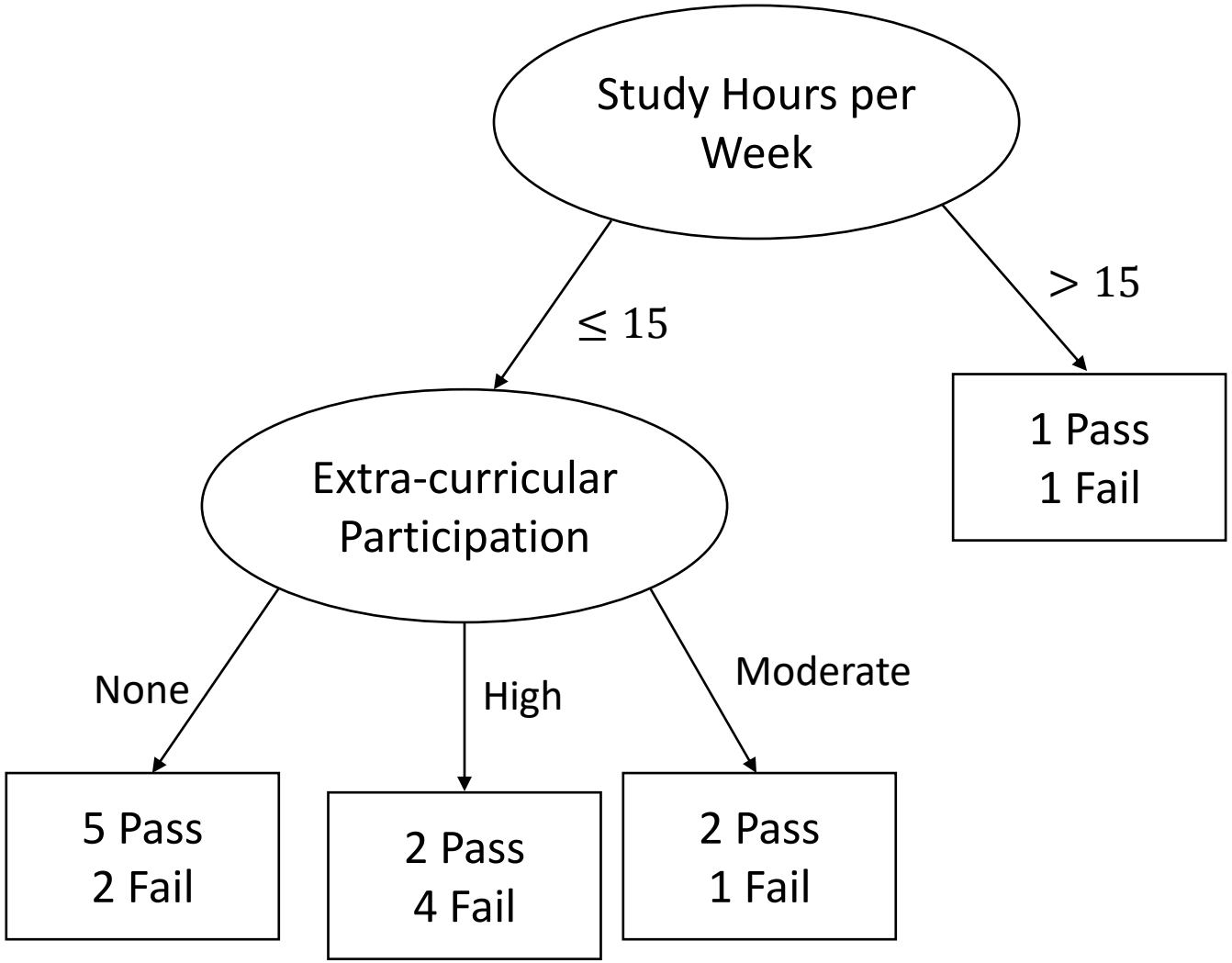
E=Number of misclassified training examples at that node

N=Total number of training examples at that node

- Step 3: Determine Estimated error using confidence intervals

- Step 4: Compare Subtree Estimated Error vs. Leaf Estimated Error
  - If Error(leaf) ≤ Error(subtree), replace subtree with leaf.

Example of Pruning in C4.5



Example of Pruning in C4.5 - Computation

- Step 1: Compute Errors
  - Error(None), Error(High), Error(Moderate)
  - Error(Extra-Curricular Participation)

- Step 2: Determine Estimated Error using confidence intervals:

$$Estimated\ Error = Observed\ Error + Confidence\ Adjustment$$

- Step 3: Compare Errors Before and After Pruning
  - If pruning increases error, keep subtree.
  - If pruning reduces error, prune subtree.

Introduction to Tree Ensembles

- What are Tree Ensembles?
  - Combination of multiple decision trees for improved accuracy.
  - Reduces variance and prevents overfitting.

- Why use ensembles?
  - Decision trees are highly sensitive to small changes in data.
  - Ensemble methods increase robustness.

Decision Trees and Their Sensitivity

- Decision Trees are highly sensitive to small data changes.

- Example:
  - Small variation in a dataset can change tree structure completely.
  - Leads to high variance in predictions.

- Solution: Use multiple trees (Tree Ensembles).

How Tree Ensembles Work

- Basic idea: Train multiple decision trees and combine their predictions.

- Each tree gives a prediction → Majority voting (classification) or averaging (regression) is used.

Bagging and Random Forests

- Bagging (Bootstrap Aggregating):
  - Sample with replacement to create multiple datasets.
  - Train a separate decision tree on each dataset.
  - Final prediction: Majority vote (classification) / Average (regression).

- Random Forest = Bagging + Feature Selection
  - Each tree selects a random subset of features at each split.
  - Reduces correlation among trees, improving generalization.

Random Forest Algorithm

- Given training set of size m, create B bootstrap samples.
- Train B decision trees, each on a different sample.
- At each split, select a random subset of features instead of all features. $k = \sqrt{n}$
- Aggregate predictions from all trees:
  - Classification: Majority voting.
  - Regression: Averaging outputs.

Boosting and XGBoost

- Boosting: Sequentially improve predictions by focusing on misclassified examples.

- XGBoost (Extreme Gradient Boosting):
  - Advanced implementation of boosting.

- Key Features:
  - Fast and efficient implementation.
  - Built-in regularization to prevent overfitting.
  - Automatically selects the best splits and stopping points.

XGBoost Algorithm

- Train the first decision tree.
- Identify misclassified samples and assign higher weights.
- Train a new tree to correct the errors of the previous one.
- Repeat for B iterations, each improving upon the last.
- Final prediction is the weighted sum of all trees.

XGBoost Example

Student	Study Hours	Attendance (%)	Pass (1) or Fail (0)
A	1	50	0
B	5	80	1
C	6	85	1
D	2	40	1
E	3	60	0
F	8	90	1

Questions?